

Vehicle Number Plate Detection

Abstract

The project aims to develop an innovative system for the accurate detection and recognition of vehicle number plates through advanced image processing and Optical Character Recognition (OCR) techniques.

With the proliferation of vehicles in urban environments, efficient management of traffic flow, law enforcement, and security surveillance has become paramount.

The proposed system offers a comprehensive solution by leveraging state-of-the-art algorithms to preprocess images, detect contours, and extract alphanumeric characters from vehicle number plates with high precision and reliability.

This project addresses the growing need for automated systems capable of seamlessly integrating into existing infrastructure to enhance traffic management, public safety, and operational efficiency.

By harnessing the power of machine learning and computer vision, the system aims to surpass the limitations of traditional Automatic Number Plate Recognition (ANPR) systems, offering improved accuracy, cost-effectiveness, and scalability.

Through modular design and open architecture, the system can be customized and deployed across diverse applications, from toll collection and parking management to security surveillance and access control.

With a focus on real-world applicability and robust performance, through improvement in accuracy, speed, image processing and other factors.

Need for the system:

The Vehicle Number Plate Detection system serves several purposes:

- **Efficient traffic monitoring:** In urban areas, efficient traffic monitoring is crucial for managing congestion, enforcing traffic laws, and ensuring public safety. Automated number plate detection can streamline this process by providing real-time data on vehicle movement and compliance.
- **Enhanced security:** Automated identification of vehicle number plates can aid in law enforcement activities, such as identifying stolen vehicles, tracking suspects, and managing access control in secure areas.
- **Streamlined processes:** Automating the recognition of vehicle number plates can simplify various processes such as toll collection, parking management, and automated entry/exit systems in gated communities or parking lots.

Existing System:

Automatic Number Plate Recognition (ANPR) systems: These systems utilize a combination of cameras, image processing algorithms, and OCR technologies to automatically detect and recognize vehicle number plates.

They are widely used in traffic management, surveillance, toll collection, and parking management applications.

Problems in the existing system:

1. **High cost:** Some ANPR systems require specialized hardware and software components, which can be expensive to procure, install, and maintain, particularly for smaller organizations or municipalities.

2. **Dependency on hardware:** Many existing ANPR systems rely heavily on dedicated hardware components, making them less flexible and scalable for deployment in diverse environments or integration with existing infrastructure.
3. **Limited accuracy:** Existing ANPR systems may struggle with accuracy, especially in challenging lighting conditions, low-resolution images, or when vehicle plates are partially obscured.

Proposed system:

The proposed system integrates state-of-the-art image processing techniques with OCR technology to address the limitations of existing ANPR systems.

The setup consists of a Raspberry Pi 3B+, running on Raspbian OS with Python 3.5, along with a Picamera attached to it.

It uses powerful Python libraries like *PIL* and free to use professional APIs like *OCRSpace* for image processing and character recognition.

The set up would first capture a video of 3 seconds of the vehicle either stationary or moving with the help of the camera module.

Then it splits the video into frames and selects the most recent image to implement further steps.

First, we *enhance the contrast* in the image to clearly distinguish the difference in intensities of the light and dark areas of the image.

This step is followed by *increasing the sharpness* of the image, to make its features more distinct and pronounced.

Then, we apply a *Gaussian Blur* to reduce the noise in the image.

Now, we convert the image into *grayscale*, making it easier to read and lightweight for processing.

We apply another *Gaussian Blur* to remove any further noise present within the grayscale image.

This is followed by *thresholding* and *drawing contours* on the Region Of Interest (ROI) in the image, i.e. the number plate.

Lastly, we perform *text recognition* using OCR methods provided by the OCRSpace API which uses *CNN algorithm* to display the final result.

After executing the project, we found that the system was able to accurately detect vehicle numbers of vehicles travelling up to 30 km/hr.

Advantages:

- 1. Improved accuracy:** Enhanced image pre-processing and OCR techniques result in higher accuracy and reliability in detecting and recognizing vehicle number plates.
- 2. Cost-effectiveness:** Utilization of off-the-shelf hardware and open-source software reduces the upfront costs associated with system deployment and maintenance.
- 3. Flexibility:** Modular software architecture and standardized interfaces enable easy customization, scalability, and integration with existing infrastructure.
- 4. Efficiency:** Automated number plate detection and recognition streamline processes, leading to increased operational efficiency and reduced reliance on manual intervention.

Features of the system:

- 1. Image pre-processing:** Contrast enhancement, sharpening, noise reduction, and other techniques to improve image quality and enhance number plate visibility.
- 2. Contour detection:** Detection of contours to isolate vehicle number plates from the background and surrounding objects.
- 3. OCR technology:** Optical Character Recognition to extract alphanumeric characters from the detected number plates accurately.

4. Scalability: Ability to scale the system for deployment in various environments and applications, from traffic management to security surveillance.

5. Customization: Flexibility to customize the system according to specific requirements and integrate it with existing infrastructure seamlessly.

6. Speed Improvement: Image processing and image capturing speed improved using OCRSpace API.

Vehicle Number Plate Detection Test Cases

Id	Test case name	Description	Input	Expected output	Actual Output	Pass / Failure
T001	Image Capture Test	Check if the camera is able to capture image from the video	Clear video of number plate of 5 seconds duration	Most recent image from video file	Most recent image from video file	Pass
T002	Pre-process Image Test	Verify if all the preprocessing steps have occurred	Most recent Image of vehicle	Greyscale image with minimum noise	Greyscale image with minimum noise	Pass
T003	Contour Cropping Test	Draw contours on the preprocessed image and crop the region of interest ROI	Pre-processed image with minimum noise	Cropped image containing area enclosed within contours	Cropped image of area bound by contours	Pass
T005	OCR Test	Validate system's ability to perform OCR on processed image	Cropped image containing area enclosed within contours	Vehicle's Correct number	Correct Number	Pass
T004	Performance Test	Decrease overall running time of program	Video containing vehicle number plate	Vehicle's correct number within 1 minute	Vehicle's correct number after 2 minutes	Fail
T005	Robustness Test	Verify if the program works for a wide range of input images	Sample Video input of various vehicles with number plate	Correct vehicle number for 75% input	Correct vehicle number for 82% input	Pass

Results



Fig 1.0: Model's ability to highlight number plate



Fig 2.0: Model's accuracy



Fig 3.0: Model's precision

Future Scope

Apart from Vehicle Number Plate Detection, we can also use the same setup for providing solutions to problems faced in the fields of agriculture and healthcare.

We can deploy the Raspberry Pi along with the camera module to solve problems like:

Crop Disease Detection, which takes input images of the crops and uses machine learning algorithms to predict which disease is ailing it.

Medical Imaging Analysis to analyse and draw preliminary conclusions from Medical images such as CT scans, X rays, etc in the field of Healthcare.

Conclusion

From this project, we have learnt about various computer vision techniques, deep learning and machine learning algorithms, and also how to configure IoT Systems to automate menial activities like Vehicle Number Plate Detection.

We have also learnt how we can use the existing setup and apply it to other fields, such as agriculture and healthcare, and how we can use Computer Vision to solve real-time problems.

Application for copyright certification has been submitted for this project

References

- <https://medium.com/@alexey.yeryomenko/automatic-number-plate-recognition-with-raspberry-pi-e1ac8a804c79>
- <https://towardsdatascience.com/i-built-a-diy-license-plate-reader-with-a-raspberry-pi-and-machine-learning-7e428d3c7401?gi=d8ab0da6fec3>
- <https://circuitdigest.com/microcontroller-projects/license-plate-recognition-using-raspberry-pi-and-opencv>
- <https://iotdesignpro.com/projects/real-time-license-plate-recognition-using-raspberry-pi-and-python>
- https://www.youtube.com/watch?v=o7_80dipEuI&t=41s
- <https://www.youtube.com/watch?v=sO7uTTq9ee4&t=820s>
- <https://ocr.space/>